Investigating ultra-local politics and demographics in Copenhagen and Frederiksberg

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I. Introduction

This project broadly investigates ultra-local politics and demographics in Denmark's capital region by analysing data from municipal elections from 2009 to 2025. We address four main areas with different data mining techniques:

A. Predicting the election

1) Which demographic features best predict vote shares for major parties in Copenhagen and Frederiksberg, and can these patterns forecast the 2025 election winner?
2) Can the number of Instagram followers and posts alone predict how many votes a party or politician gets?

B. Swing states

1) Which polling areas in Copenhagen and Frederiksberg municipalities are "swing states"?
2) What are the factors deciding if a polling area is a "swing state" or not?

C. Characterising neighbourhoods

1) How do neighbourhoods in Copenhagen and Frederiksberg cluster based on demographic and socioeconomic characteristics, and what distinguishes these clusters?
2) How can outliers, if they exist, be explained?

D. Foreign voters and migration

1) How does population composition by citizenship status affect voter turnout?

II. Data and limitations

Our analyses primarily use data from Den Danske Valgdatabase, including voting results from previous municipal elections (2009, 2013, 2017, 2021) and demographic data, and new voting data on the 2025 election results from valg.dk. The unit of observation is polling areas; across the four chosen historical election years, there are around 58 such areas across Copenhagen and Frederiksberg municipalities. We also use scraped Instagram data, migration data from Danmarks Statistik, geodata on polling area borders from DAGI, and party letter mappings from valg.dk.

The data have some limitations. Polling areas are small, administrative geographical areas created for elections and not for reflecting demographic or socioeconomic patterns. In addition, polling areas often change, which complicates temporal comparisons. Den Danske Valgdatabase deals with this by merging some polling areas to enable cross-year comparisons. Because of this, and because of Kommunalreformen in 2007, which changed a lot of administrative borders, we decided to only use data from 2009 onwards. In addition, data from Den Danske Valgdatabase is very poorly documented. This means that we can't confirm the timeliness for demographic features, and we have to interpret their meaning based on the variable name. Finally, the raw demographic data alone contains more than 13,000 columns across 16 CSV files. In other words, we're dealing with very high-dimensional data, which require extensive preprocessing.

III. Initial preprocessing

Entity resolution and schema integration were particularly important in the initial preprocessing, as we consolidated data from multiple sources, files and formats. So was data reduction.

For demographic data, we manually selected the most interesting feature categories, which contained 12 919 variables. We iteratively reduced this number with binning and data aggregation. Duplicate columns were removed, and columns with all NaN values were dropped. For columns with some zero values, these were kept as legitimate values (e.g., a polling area might genuinely have zero citizens from former Yugoslavia). We eventually integrated data based on polling area identifiers. At the end of this process, we had around 100 demographic variables for each election year. Absolute counts were converted to percentages based on population, while median household income and gross top 20% income were kept as-is since they're already relative measures.

Historical election data was transformed with schema integration, converting to numeric types, splitting by year, adding geography columns and removing irrelevant ones, translating Danish column names, and calculating relative vote shares (percentage). The 2025 election data had a different structure altogether and required concatenating separate Copenhagen and Frederiksberg files.

One challenge was inconsistent party letter codes in raw election data: The same letter can represent different parties in different years or municipalities, and the same party can use different letters over time. But vote columns are just labeled with letters, which creates ambiguity when combining years or municipalities. To deal with this, we manually created year-specific and municipality-specific mappings by reviewing valg.dk for each election and documenting correspondences in a CSV file.

Geodata, which we later used with geojson.io to understand the polling areas and their data better, was converted from regular JSON to GeoJSON and filtered to Copenhagen and Frederiksberg.

Instagram data was collected using a custom scraping pipeline implemented with Playwright. The scraper reads a CSV file containing Instagram handles, accesses each public profile, and extracts follower counts and the number of posts directly from the profile page. Authentication is handled through a reusable session file to ensure consistency across runs. Scraped values are written to a new CSV file, preserving all existing metadata and row order, which allows for reproducible data collection. This approach was applied to both parties' and candidates' Instagram accounts.

IV. Predicting the election

A. Preprocessing and Feature Selection

After the general preprocessing described earlier in the report, further preprocessing was required for this research field. The main issue was reducing the number of features. This resulted in firstly aggregating features regarding age, education level, car ownership, and benefit type, by putting them into broader categories. During this process we also dropped features which we deemed too specific for the analysis. This included dropping all citizenship features, specific immigrant features (only keeping the total amount of immigrants for an area), counts of specific housing types and when the housing was built. We chose to keep all socioeconomic features and all income features. As part of our general preprocessing the data had already been converted into proportions instead of absolute numbers (i.e., normalized by population), however this did not include the median household feature. For this research question we chose to normalize median household income across all four historical election years (2009, 2013, 2017, 2021) using a log-transformation followed by min-max scaling to a 0-100 range to match it to the other demographic features, spanning all years to ensure consistent scaling. After this round of preprocessing we had 22 demographic features.

1) Correlation Analysis: To investigate possible features to drop, we performed correlation analysis for each historical election year. This entailed making a correlation matrix and setting a threshold of 0.80 correlation to identify the most correlated features in order to see if one of the features could be dropped. The correlation analysis showed in general that many of our demographic features are correlated, however this is expected for this type of data. Three highly correlated feature pairs did stand out since they had almost perfect correlation across the election years and covered the same underlying tendencies. These where Seniors_65_plus and Benefit type_Folkepension (corr: 0.997-0.999), Socioeconomy_Top executives and Income_Gross income top20pct (corr: 0.91-0.93) and Housing tenure_People_Ejer and Households_with_car (corr: 0.81-0.89). The latter of each pair was dropped from the data. For the remaining highly correlated features, it was decided to keep them in the data since their correlation is most likely caused by underlying societal trends or they are the inverse of each other. This brings our demographic feature count down to 19 features.

2) Regression-Based Feature Selection: Since we aim to answer the question "Which demographic features best predict vote shares for major parties in Copenhagen and Frederiksberg" we wanted to limit our features even further. Therefore we decided to use ElasticNet regression for further feature selection with cross-validation to predict vote shares for each of the five major parties (A, V, C, F, Ø) separately. Our strategy for the feature selection was to use the cross-party validation to ensure that features continuously held importance across multiple parties since they are then more likely to represent robust, generalizable predictors.

The ElasticNet regression was applied with the following parameters:

- L1 ratios: [0.1, 0.3, 0.5, 0.7, 0.9] to balance Lasso and Ridge regularization
- 5-fold cross-validation to optimize regularization strength
- Maximum 20,000 iterations to ensure convergence
- Features standardized prior to modeling to ensure equal weighting

Two complementary analyses were conducted: a temporal analysis to assess feature stability over time and a pooled analysis to identify overall predictive features. The features were ranked by coefficient magnitude, cross-party consistency and temporal stability.

3) Final Feature Selection: Based on cross-party importance, temporal stability, and coefficient magnitudes, ten demographic features were selected for final modeling (Table I).

TABLE I
FINAL SELECTED DEMOGRAPHIC FEATURES

Rank	Feature	Mean Importance
1	People_on_benefits	0.797
2	Seniors_65_plus	0.733
3	Education_Long_higher_education	0.731
4	Socioeconomy_Unemployed	0.717
5	Socioeconomy_Top executives	0.667
6	Housing type_Apartment buildings	0.614
7	Education_Primary_lower_secondary	0.586
8	Housing tenure_People_Ejer	0.564
9	Education_Vocational_training	0.375
10	Housing tenure_People_Almen	0.347

B. Predicting the Election

This section presents the results of using demographic features to predict municipal election outcomes in Copenhagen and Frederiksberg polling areas. The analysis reveals significant limitations in demographic-only prediction models when applied across periods of political realignment. To do this we used the same ElasticNet regression set-up as in the feature selection.

Following the feature selection, the ten demographic features (mentioned in Table I) were retained for the prediction.

We chose to have three validation approaches to assess the model performance. Since we did not have demographic features aligned with the 2025 election, we wanted to do a temporal lag validation, where we tested how well demographics from one election year could predict for the next election year. We then use traditional holdout validation by using 2009-2017 for training and 2021 for testing. Lastly we use the 2021 demographics as a proxy for the 2025 demographics to try and predict the 2025 election results.

To evaluate we focus on the metrics Mean Absolute Error and R^2.

1) Temporal Lag Validation: The temporal lag validation assessed whether using older demographic data (from 4 years prior) could predict future elections, directly testing the viability of using 2021 demographics to predict the 2025 election.

TABLE II
TEMPORAL LAG VALIDATION RESULTS

Test Period	Average MAE	Average R^2
2009→2013	6.32%	-7.047
2013→2017	3.30%	-0.209
2017→2021	5.25%	-0.705
Overall	**4.96%**	**-2.654**

From this we find that the overall average MAE is acceptable at 4.96%, but not very good. However, the negative R^2 values across all periods indicate that the predictions perform worse than simply predicting the mean vote share, which did not look promising for our use of the 2021 demographic features as a proxy for the 2025 demographic data.

2) Same-Year Validation (2021 Holdout): Applying traditional validation using 2009-2017 data to predict 2021 outcomes (using actual 2021 demographics) revealed fundamental challenges in the model's generalizability (Table III).

TABLE III
2021 HOLDOUT VALIDATION PERFORMANCE BY PARTY

Party	MAE	R^2
A	7.11%	-2.135
V	2.63%	-0.628
C	4.95%	0.583
F	1.94%	-0.018
Ø	7.60%	-0.624

Here it shows that only Party C achieved positive R^2 (0.583), indicating some predictive power. All remaining parties showed negative R^2 meaning as we saw with the temporal lag validation that the model performs worse than baseline predictions. This is also evident in the accuracy of 44.8% which not remarkable. These results suggest that 2021 represents a departure from the demographic-voting patterns established in 2009-2017.

3) Cross-Validation Performance on Training Data: Despite poor holdout performance, cross-validation on the full 2009-2021 dataset showed strong within-sample predictive power (Table IV).

TABLE IV
CROSS-VALIDATION PERFORMANCE ON TRAINING DATA

Party	Cross-Validated R^2
A	0.555
V	0.400
C	0.623
F	0.874
Ø	0.735

This discrepancy (i.e., high within-sample R^2 but negative out-of-sample R^2), indicates that the model successfully learns patterns in the training data but these patterns do not generalize to prediction periods, particularly 2021.

4) 2025 Prediction: Our final predictions for the 2025 election results using the 2021 demographic data as a proxy yielded the following results (Table V):

TABLE V
2025 ELECTION PREDICTIONS

Party	Pred. Mean	Actual Mean	MAE	R^2
A	22.8%	13.9%	9.01%	-5.758
V	8.1%	5.6%	2.63%	-1.804
C	14.8%	13.7%	3.62%	0.737
F	10.7%	16.6%	6.22%	-2.189
Ø	20.8%	22.0%	2.99%	0.672

Again most parties showed negative R^2. Here only Parties C and Ø showed positive R^2 (0.737 and 0.672 respectively).

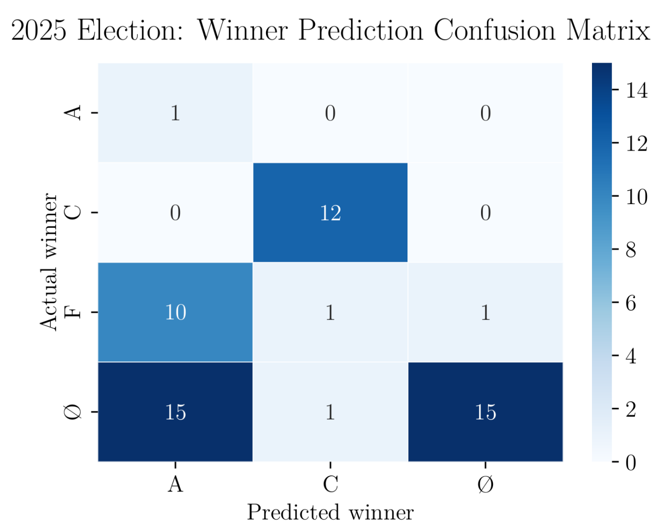

Fig. 1. Confusion matrix of the 2025 prediction.

The model severely over-predicted Party A wins and it completely failed to predict any Party F wins as can be seen on Figure 1.

The winner prediction accuracy is at 50.0% which is a slight improvement from the prediction of the 2021 election.

Overall this again tells us that the model using the selected demographic features fails to predict the 2025 election adequately.

5) The Failure of Demographic Prediction: The results demonstrate a fundamental limitation of demographic-only electoral prediction models when applied across periods of political realignment. The strong cross-validation performance (R^2 = 0.40-0.87) contrasts sharply with poor holdout performance (R^2 = -5.76 to 0.74), indicating that demographic features successfully predict voting patterns *within* stable political periods but fail to generalize *across* periods of political change. Party A's collapse from 43-44 wins (2009-2017) to 1 win (2025) reveals a political shift that demographic features cannot capture, likely driven by factors such as political trends, campaign effects, or issue salience.

C. Conclusion

The analysis demonstrates that demographic features, while predictive within stable political periods (R^2 = 0.40-0.87 on 2009-2021 data), fail to predict elections across periods of political realignment (R^2 = -5.76 to 0.74 on 2021-2025 predictions). Party A's collapse from 43-44 wins (2009-2017) to 1 win (2025), and Party Ø's surge to 31 wins, represents a political shift that demographic features cannot capture.

This negative result is useful since it demonstrates the boundaries of demographic-based prediction and highlights the importance of incorporating temporal dynamics. The proposed investigation of temporal electoral features represents a promising avenue for improving prediction by capturing things such as voting behavior.

Given the limitations of demographic-only models, future work should investigate temporal electoral features (variables capturing voting history and patterns of change) which may prove more robust to political realignments by capturing voting behavior itself rather than just voter characteristics. This could include testing both temporal features alone and hybrid models combining demographic and temporal features.

D. Instagram

The analysis combines official vote counts with Instagram-derived metrics. While parties and candidates differ substantially in scale, they were analysed separately using methods appropriate to each, with the shared goal of assessing the explanatory and predictive value of Instagram presence.

The party-level dataset consists of 28 political parties. Those without an Instagram presence were retained, with missing follower and post counts set to zero. A binary indicator variable was introduced to distinguish parties with and without an Instagram account, allowing the analysis to separate the effect of social media presence from the scale of activity. Due to a strong right skew in vote counts and Instagram metrics, log-transformations were applied using a $\log(v + 1)$ specification, which preserves zero values.

Exploratory analysis at the party level revealed moderately strong correlations between Instagram metrics and vote counts. To evaluate whether these associations translated into predictive power, several linear regression models were estimated, of which the best-performing specification was a log–log model including a binary indicator for Instagram presence. This model achieved an R^2 of approximately 0.85 when estimated on all parties, suggesting substantial explanatory power within the observed data. However, the model proved highly sensitive to extreme observations. Removing the top five percent of parties caused cross-validated R^2 to drop to strongly negative values - this suggests that the model's predictive power is driven by a few highly visible parties rather than a stable relationship between Instagram activity and electoral outcomes.

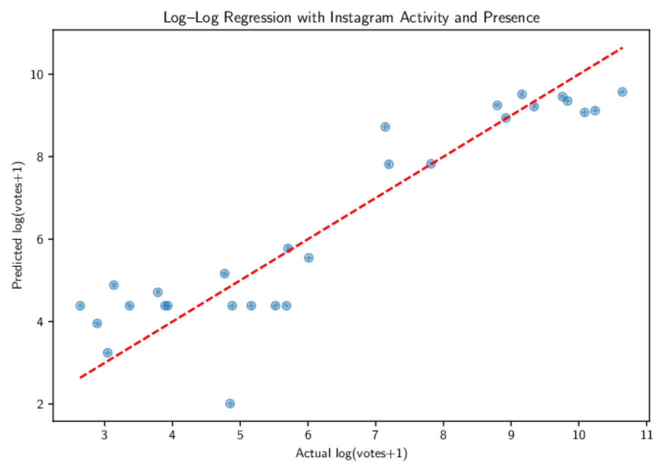

Fig. 2. While the model captures a general upward trend between actual and predicted votes, the deviations from the 1:1 line indicate that Instagram activity provides insufficient and inconsistent information to reliably predict electoral outcomes.

In contrast, the candidate-level analysis involved a dataset of 410 individual candidates. This larger and more diverse data sample required a different modeling approach. As with parties, missing Instagram data was handled by introducing binary indicators and replacing missing values with zero. Log-transformations and interaction-based features were again applied to address skewness and capture the combined effects of reach and activity. Party affiliation and municipality were included as categorical control variables.

Exploratory analysis for candidates showed a positive but noisy relationship between follower count and votes, while posting frequency exhibited a weaker association. Given the evident non-linearities in the data, a Random Forest regression model was used to analyse candidate-level outcomes. The model achieved moderate predictive performance, explaining approximately 28% of the variance in candidate vote counts. Diagnostic plots indicated that the model captured relative candidate ranking reasonably well but systematically overestimated high-performing candidates and underestimated low-performing ones. Feature importance analysis consistently identified log-transformed follower count as the most influential predictor.

In conclusion, Instagram presence and follower count are

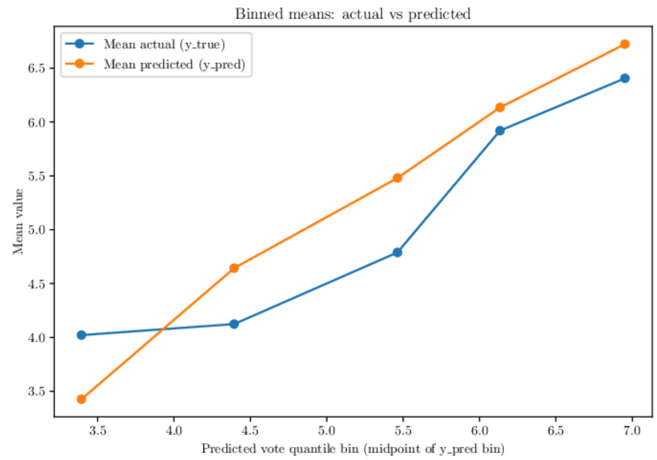

Fig. 3. Binned comparison of predicted and observed candidate vote counts for the Random Forest model.

positively associated with electoral outcomes, but they do not provide a stable or robust basis for prediction. At the party level, predictive performance is highly sensitive to extreme observations. At the candidate level, non-linear models capture meaningful patterns but still leave the majority of variance unexplained. These results suggest that Instagram metrics primarily reflect visibility rather than acting as determinants of electoral success.

V. Swing States

This section identifies ”swing states” among Copenhagen and Frederiksberg’s polling areas and examines the demographic and electoral factors that distinguish competitive areas from stable ones.

1) Feature Engineering and Swing Labels: Electoral competition measures (margin, Herfindahl index, party switches) directly capture swing behavior through close races and changing winners. Demographic features (age, education, socioeconomic status, housing, citizenship) proxy for partisan alignment and political volatility. These indicators, averaged across 2009-2021, define three swing labels:

1) Switch-based measure: Areas where the winning party changed at least once across elections
2) Low-margin measure: Areas where the average margin between the top two parties was below 10 percentage points
3) Combined measure: Areas meeting both criteria

2) Model Selection and Performance: To avoid high dimensionality, we trained Random Forest classifiers for each swing definition, retaining the 20 most important features via impurity-based selection for final modeling and SHAP analysis. As shown in Table VIII, all three models performed well, each achieving a perfect ROC-AUC score of 1.000. The margin-based classifier achieves perfect test performance, outperforming a naive baseline that would classify all areas as the majority class, confirming that demographic and electoral features provide meaningful signal for swing state identification. Given its strong generalization and perfect test results, we chose the margin-based definition for the final analysis.

TABLE VI
Performance Comparison of Swing State Definition Methods

Method	Swing States	Test F1	ROC-AUC	Overfitting Gap
Switches	37	0.933	1.000	0.040
Margin	35	**1.000**	1.000	-0.022
Combined	30	0.923	1.000	0.062

A. Geographic Distribution

Of 58 polling areas, 35 (60.3%) are swing states, concentrated entirely in Copenhagen (Table VII). Frederiksberg’s complete absence of swing states suggests distinct political dynamics between municipalities.

TABLE VII
Swing States by Municipality

Municipality	Total Areas	Swing States	Percentage
København	51	35	68.6%
Frederiksberg	7	0	0.0%
Total	58	35	60.3%

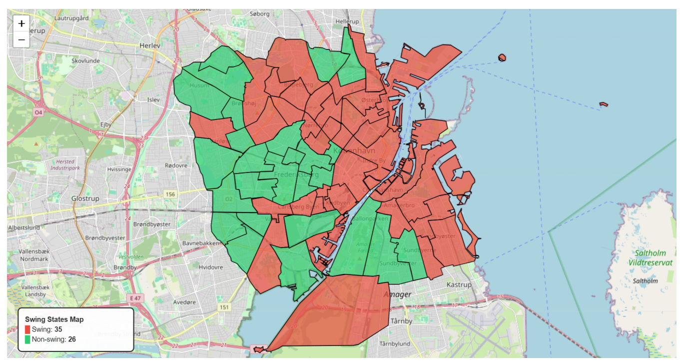

Fig. 4. Map of Copenhagen showing electoral swing status by district, with swing constituencies in red and non-swing constituencies in green.

Figure 4 shows the geographic distribution, revealing that swing states are distributed across Copenhagen but with concentrations in certain areas. The swing areas include parts of the city from central districts to outer areas, suggesting that ”swing” politics is not only based on specific geographic zones. This geographic pattern connects to the neighborhood clustering analysis in Section VI, where we identified five distinct demographic clusters. Swing areas show up in lots of different kinds of neighborhoods, which suggests that electoral volatility isn’t tied to one specific “type” of place but can grow out of many different demographic mixes.

B. Demographic Characteristics of Swing States

SHAP analysis (Appendix A + Table VIII) show the demographic factors distinguishing between swing and non-swing areas. The strongest factor is the share of young adults (ages 18–39). Swing states have an average of 46.6% in this age group, compared to 38.6% in non-swing areas. Conversely, non-swing areas have higher elderly populations (13.8% vs. 9.8%), suggesting older voters maintain more stable partisan loyalties.

Cooperative housing ("Andel") shows the largest absolute difference, with 15.3% higher in swing states, where apartment buildings also have higher prevalence. These patterns likely reflect the younger demographic and urban lifestyle characteristics of swing areas.

TABLE VIII
COMPARISON OF SWING AND NON-SWING CONSTITUENCIES

Feature	Swing Mean	Non-Swing Mean	Difference
Age 18–39 (%)	46.6	38.6	+8.0
Age 65+ (%)	9.8	13.8	-4.1
Andel Housing (%)	41.7	26.4	+15.3
Apartment Buildings (%)	88.6	81.4	+7.2
Long Higher Education (%)	27.0	21.8	+5.3
Primary/Lower Secondary (%)	16.8	20.8	-4.0
Voter Turnout (%)	60.6	61.0	-0.4
Top 2 Margin (pp)	7.7	15.1	-7.4
Herfindahl Index	0.163	0.192	-0.029
Effective # Parties	6.2	5.4	+0.8

Citizenship patterns reveal additional distinctions: higher South and Central American populations associate with swing behavior, while Old EU citizenship tilts toward stability. Meanwhile, North American citizenship has a more mixed influence, both high and low proportions seem to push areas toward swing classification. Overall, this suggests that swing states are not only defined by younger populations, but also by migration profiles, possibly linked to different political leanings or weaker political commitment than those seen in more established immigrant communities.

Education levels show nuanced patterns, with a higher amount of people with long higher education in swing areas and a higher amount of people with primary/lower secondary education in non-swing areas. This suggests that swing areas contain more highly educated people, but could also correlate with more children (0-17) living in those areas.

C. Party Competition Patterns

Party A dominates both area types, but faces different challengers: Party Ø in swing areas versus Party C in non-swing areas (Figure 5), reflecting Frederiksberg's strong C preference.

In swing areas, the main fight is between two left-wing parties (A vs Ø), while in more stable areas it is between a center-left and a center-right party (A vs C). This suggests that swings are more likely when parties are ideologically close, because voters can more easily move between options that still match their core views. The party switches between elections (Figure 5, right panel) happen primarily between parties that sit close to each other on the political spectrum, with the A-Ø switches especially common in the swing areas, strengthening this pattern.

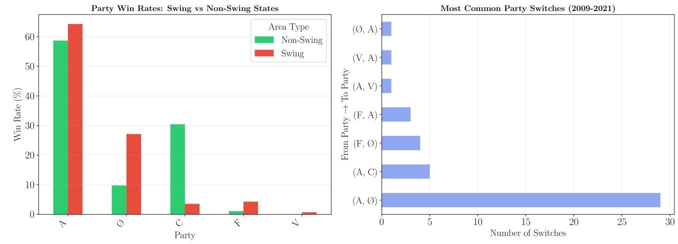

Fig. 5. The figure displays two panels: the left panel shows bar charts of party win rates in swing and non-swing areas, while the right panel shows a horizontal bar chart of the most common party-to-party switches between elections from 2009 to 2021.

D. Connection to Predicting The Election

The swing-state analysis helps make sense of why the demographic prediction model broke down in Section IV. The model looked great in cross-validation, but its real-world performance, especially for the 2025 election (R^2 from -5.76 to 0.74), was much weaker.

Swing areas, which make up 68.6% of Copenhagen's polling districts, are exactly where demographics explain the least. These places tend to be younger, more highly educated and have more fragmented party systems, so static demographic snapshots cannot keep up with how politics shifts over time.

On top of that, swing districts are concentrated in Copenhagen, while Frederiksberg is completely stable, showing that local political culture at the municipal level really matters. Any demographic model must account for these area-specific dynamics that go beyond individual voter traits. The collapse of Party A, from 43 to 44 victories in 2009–2017 to just 1 in 2025, occurred mainly in swing areas where demographics still pointed to strong support, but short-term forces instead triggered a major change.

VI. CHARACTERISING NEIGHBOURHOODS

A. Preprocessing and exploratory data analysis

Since we will base this analysis solely on 2021 data, we do feature selection on 90 demographic features. We remove redundant and uninteresting variables, then do binning and data aggregation on age groups, which are collapsed from 5-year increments into 4 bins: 0-17 (children), 18-29 (young adults), 30-69 (working age) and 70+ (elderly). We aggregate citizenship and immigration columns to the categories Western (Nordic, old EU, North America, South and Central America), Global South (Africa, Asia and Oceania, Turkey), and Eastern Europe (new EU and former Yugoslavia); Denmark and Other Europe are kept as is.

Violin plots on selected demographic features reveal some outlier values within crime, Danish citizenship, and social housing. We keep these in the analysis rather than remove them, since the hypothesis is that these outliers are genuine characteristics of a specific neighbourhood rather than data errors.

The first cluster analyses show that using both demographic and voting data clusters nearly all Frederiksberg areas separately from Copenhagen areas, regardless of demographics and socioeconomics. Since Copenhagen and Frederiksberg hold separate municipal elections with different candidates and local issues, this seems to reflect local politics rather than neighbourhood characteristics. Because of this, we exclude the voting data from the analysis. The number of dimensions have thus been reduced to 45.

B. Models

1) Standardisation: All variables are standardised using z-score scaling before clustering. Our variables have very different scales: Most are percentages, but the income variables are much higher. Without standardisation, these would dominate distance calculations. Z-score standardisation transforms each variable to mean=0 and std dev=1: $z = \frac{x-\mu}{\sigma}$.

2) Dimensionality reduction: We apply Principal Component Analysis (PCA) to the standardised dataset to reduce dimensionality while retaining the majority of variance in the data. PCA creates new orthogonal variables (principal components) that are linear combinations of the original variables, ordered by the amount of variance they explain. Cumulative explained variance indicates that the first 4-7 principal components explain between 75% and 90% of the total variance, with the "elbow point" in the scree plot also occurring somewhere between the 4th and 7th component. We ultimately choose the five principal components for the clustering analysis, as these explain just over 80% of the cumulative variance.

PCA assumes linearity and is less good at capturing non-linear structures in the data, and it is optimised to explain variance rather than to separate clusters. t-Distributed Stochastic Neighbour Embedding (t-SNE) might be better at revealing hidden clusters in the data. However, our demographic features often correlate, and PCA is useful for reducing that redundancy by creating uncorrelated components. In addition, the principal components are easier to interpret than for example two new dimensions visualised with t-SNE, and it is easier to use principal components for further analyses. Still, we use t-SNE for 2D scatter plots, which later helps confirm clusters.

3) Clustering: We first determine the optimal number of clusters k using the elbow method, which gives us $k = 5$.

Three clustering algorithms are used and compared: k-means clustering, hierarchical clustering (with Ward's linkage method), and Gaussian Mixture Models (GMM). All three are fitted to the five principal components. The algorithms are evaluated using Silhouette Score (how similar is each point to its own cluster compared to other clusters), the Davies-Bouldin Index (the average similarity ratio of each cluster with its most similar cluster, lower is better), and the Calinski-Harabasz Index (the ratio of between-cluster variance to within-cluster variance).

k-Means outperforms the other algorithms on all three metrics (see Table IX), but it is worth noting that a silhouette score of 0.304 is not very high; scores close to 1 show well-defined clusters, and scores close to -1 show misclassification. This tells us that even though we can divide polling areas in Copenhagen and Frederiksberg into clusters, the differences between the clusters might not be that big.

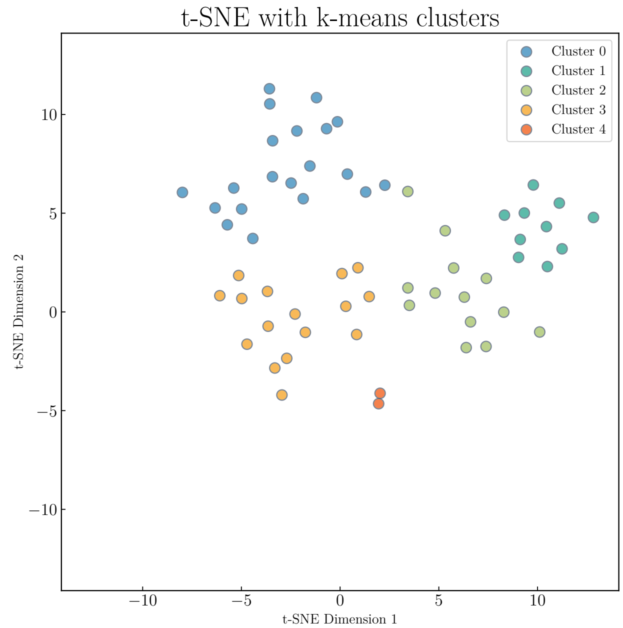

Fig. 6. t-SNE visualisation of polling areas based on all demographic features. The colours show clusters found with k-means based on 5 principal components from PCA.

TABLE IX
EVALUATION METRICS FOR CLUSTERING METHODS

Method	Silhouette score	Davies-Bouldin	Calinski-Harabasz
k-Means	0.304	0.973	22.747
Hierarchical	0.283	1.053	21.678
GMM	0.225	1.034	16.890

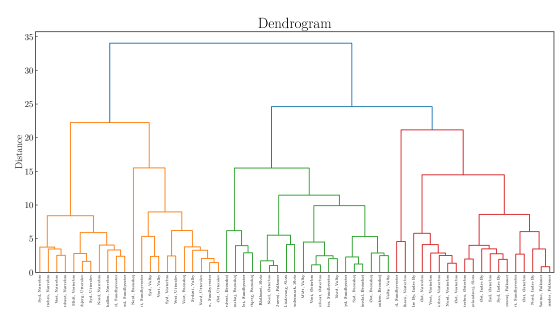

Fig. 7. blabla

4) Interpreting clusters: We calculate eta-squared values (η^2) for the original 45 features to identify which demographic features most strongly differentiate the clusters. Eta-squared provides a standardised measure of effect size, indicating the proportion of total variance in each variable that is explained by cluster membership: $\eta^2 = \frac{SS_{between}}{SS_{total}}$, where $SS_{between}$ is the sum of squares between clusters and SS_{total} is the total sum of squares. The values range from 0 to 1; higher values indicate that cluster membership explains more of the variance in that variable. We use some of the top features to visualise the clusters in heatmaps and radar charts. These reveal that education levels (particularly higher education vs. primary education), immigration and citizenship (Western vs. Global South vs. Eastern Europe), housing (type, tenure, construction period), welfare benefits, voter turnout and income best describe the differences between each cluster. We also describe each cluster by their most distinctive demographic, socioeconomic, and housing characteristics by examining how cluster means deviate from global means across all features.

5) Identifying outliers: We first apply DBSCAN (Density-Based Spatial Clustering of Applications with Noise) independently of the clustering with parameters $\epsilon = 4.2$ and `minimum points = 3` to identify polling areas in low-density regions of the feature space. Three such polling areas are found: 'Syd, Sundbyvester' (Ørestad), 'Sydhavn, Vesterbro' (new Sydhavn), and 'Nord, Brønshøj' (Tingbjerg). Interestingly, k-means with $k = 5$ assigns the first two of these to their own small cluster, suggesting that these two areas together form a cluster, but they are prevented from doing that with DBSCAN when `minimum points = 3`.

We then use k-means to identify outliers based on distance from cluster centroids. The polling areas furthest from their assigned cluster centroids are 'Indre By, Indre By', 'Nord, Brønshøj' and 'Søndermark, Slots'. These will be described in the next section.

C. Findings

We have identified 5 five clusters of polling areas. Our findings are limited to the fact that the demographics data is not newer than 2021, and possibly older; some things may have changed since then.

Cluster 0 is urban with established and affluent intellectuals and some western immigration. It's characterised by higher education levels, Western immigration, top executives and entrepreneurs, co-op housing and higher income for the top 20 % earners. It has fewer detached houses, social housing, people on welfare benefits, people from Eastern Europe and the Global South. This cluster contains most of Frederiksberg, Indre by, Christianshavn and Islands Brygge, central parts of Østerbro, some of Vesterbro, and a single polling area in Nørrebro.

The polling area 'Indre By, Indre by' (the most central area) is labeled as an outlier, because, relatively to other areas in its cluster, it has more crime (arrests), is more male-dominated, has more citizens from Western countries and less Danish citizens, and also has more people on social benefits, lower voter turnout, and more home renters.

Cluster 1 is socially vulnerable with low income and education levels and high shares of social housing, people on 'kontanthjælp' and disability pension, and people from

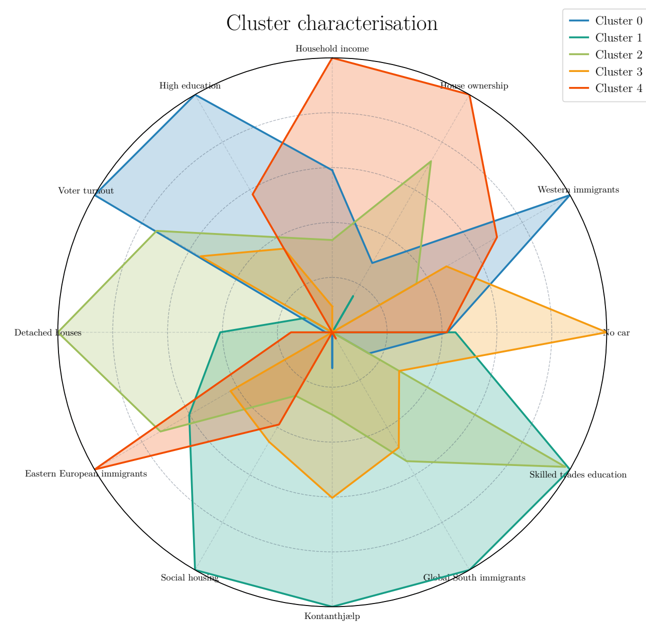

Fig. 8. Cluster 0 is urban with established and affluent intellectuals and some western immigration. Cluster 1 is socially vulnerable with low income and education levels and high shares of social housing, people on 'kontanthjælp' and disability pension, and people from the Global South and Eastern Europe. Cluster 2 is suburban with high shares of homeowners with multiple cars. Cluster 3 is urban with well-educated young adults mixed with some poverty. Cluster 4 is primarily characterised by hyper-modern apartment housing, very high income and lots of internationals.

the Global South and Eastern Europe. Western immigration is low, there's little co-op housing, fewer top executives and entrepreneurs. The cluster contain Tingbjerg, some other parts of Brønshøj and Utterslev, old Sydhavn, parts of Amager (Urbanplanen), some outer parts of Valby (Folehaven and Vigerslev). These areas form a banana-shaped arc from north-western to southern Copenhagen, reminiscent of the controversal term "Den Rådne Banan" (The Rotten Banana) once used to describe economically marginalised parts of peripheral Denmark.

'Nord, Brønshøj' (Tingbjerg) is an outlier. This is the only polling area which consists only of social housing. Here, the education levels are lower, more people are on benefits, the majority of people living here are immigrants from the Global South, the voter turnout is around 33 percent and only 50 percent have Danish citizenship. It's also a family neighbourhood with more children than other areas in the same cluster.

Cluster 2 is suburban with high shares of homeowners with multiple cars. There are high shares of detached houses, home ownership, people educated within skilled trades and households with two or more cars. There are fewer apartments and new housing, less immigration from Western countries and 'Other Europe', less young adults with long university degrees. The cluster includes most of the villa-heavy polling areas in Brønshøj, Vanløse, outer parts of Frederiksberg, and some parts of Amager.

The polling area 'Søndermark, Slots' (around Roskildevej on outer Frederiksberg) is an outlier. It's less suburban with more apartments and less villas. It has more elderly people on old-age pension and more females (likely old females who's outlived their husbands).

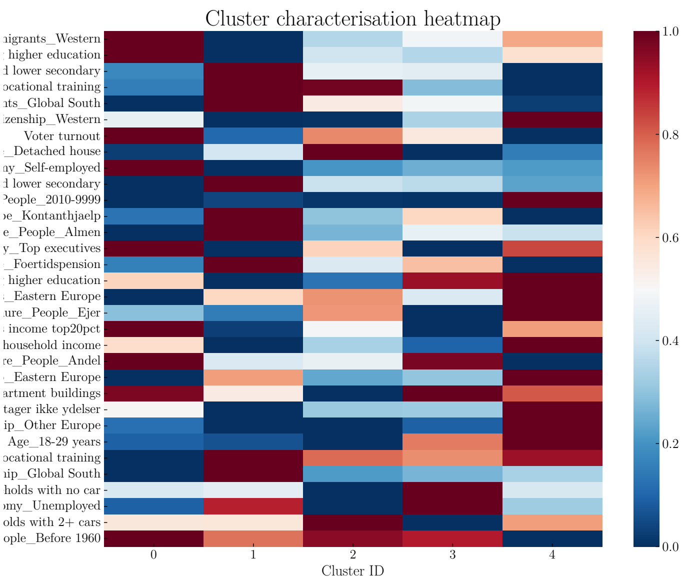

Fig. 9. Correlation between clusters and demographic variables

Cluster 3 is urban with well-educated young adults mixed with some poverty. It has high shares of young adults, co-op housing, young adults with long university degrees, and also people on 'kontanthjælp' and disability pension, which may indicate that the neighbourhoods in this cluster have not yet been completely gentrified. The cluster has fewer detached houses, fewer people that own the homes they live in, fewer top executives and newly built housing, lower income, and fewer households with cars. It contains all of Nørrebro (except for central 'Øst, Nørrebro'), Amagerbro, some of Nordvest, a few polling areas in Valby and outer Østerbro, and a single area in Vesterbro.

Cluster 4 is primarily characterised by hyper-modern apartment housing, very high income and lots of internationals. People mostly live in housing built after 2010, income is high, home ownership is high, so is immigration from Western countries and Other Europe, and there are more young adults with long university degrees. The cluster scores low old housing and co-op housing, detached houses, people on benefits, and education levels that are very low or within skilled trades.

Further work should visualise the clusters and their differences/similarities on geographical maps. It would also be interesting to merge the clustered polling areas back with the voting data to see political characteristics of each cluster, or collect data from national elections and include that in the clustering algorithms. That would make the political data more comparable across municipality borders.

VII. Foreign voters and migration

The analysis combines polling-area–level election results and population composition data across four parliamentary elections (2009, 2013, 2017, and 2021).

Exploratory analysis reveals significant diversity across polling areas. Mean voter turnout increases from around 56% in 2009 to approximately 65% in following elections. Despite this overall increase, differences in turnout across polling areas remain stable over time. Similarly, polling areas differ in their non-Danish population shares, and these differences remain largely stable in their relative ranking across elections.

Across all four elections, polling areas with higher non-Danish population shares consistently show lower voter turnout. Grouping polling areas into quintiles by non-Danish share reveals a stable turnout gradient of roughly 10–15 percentage points between the lowest and highest. Turnout levels are highly correlated across elections, indicating that differences across polling areas are driven primarily by persistent local characteristics rather than short-term electoral dynamics.

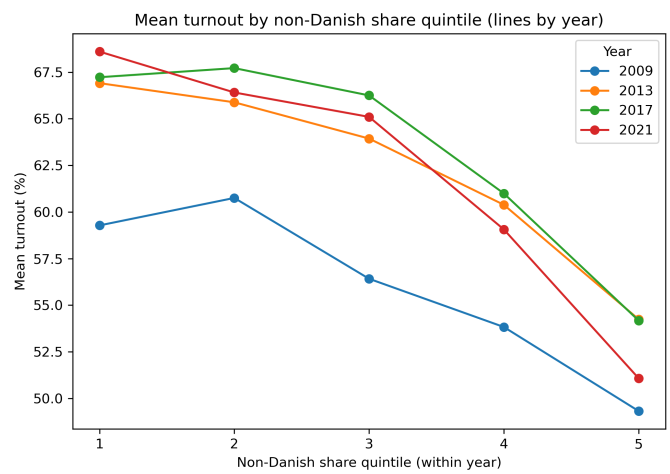

Fig. 10. Polling areas with higher non-Danish population shares consistently exhibit lower voter turnout, simultaneously, increases in overall turnout over time do not substantially alter the turnout differences between polling areas.

Party vote shares exhibit similarly stable spatial patterns. Correlations between party vote shares and non-Danish population shares are notable for several parties and remain mostly stable across elections, suggesting persistent demographic–political alignment at the polling-area level. In contrast, short-term changes in non-Danish population shares are only weakly associated with changes in turnout or party support between elections.

Taken together, these findings indicate that polling areas are characterised by stable combinations of demographic composition, turnout behaviour, and party support. This motivates the use of clustering methods to summarise long-run spatial political structure rather than focusing on short-term changes.

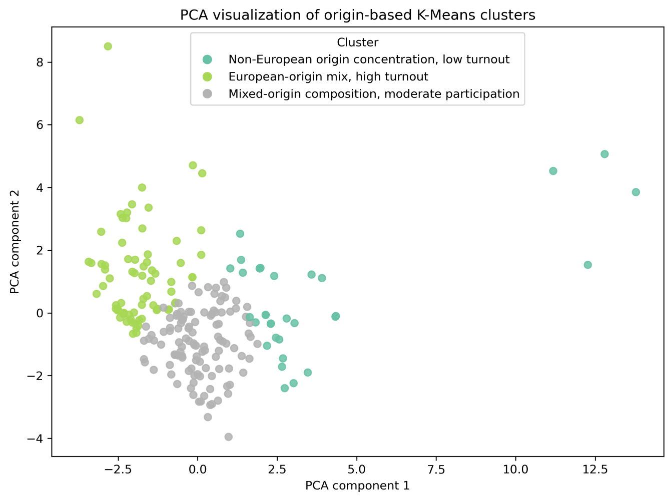

Fig. 11. PCA Component 1 primarily reflects differences in citizenship-origin composition, separating polling areas with predominantly European origins from those with higher shares of non-European origins. PCA Component 2 captures secondary variation related to political participation and party system characteristics, including voter turnout and party fragmentation.

K-Means clustering is applied to identify groups of polling areas with similar profiles in terms of migration composition. Rather than clustering on individual party vote shares, political outcomes are summarised using party dominance and party fragmentation measures to reduce dimensionality and allow comparison across elections. Migration composition is represented using regional origin shares, excluding the Danish category to avoid redundancy. All variables are standardised prior to clustering. The number of clusters is selected using a combination of elbow analysis and silhouette scores. While solutions with two to five clusters were examined, a three-cluster solution provided the best balance between cluster separation and interpretability, avoiding both over-fragmentation and excessive aggregation.

The resulting clusters are distinct:

1) **Low-turnout, non-European-origin cluster.** This cluster comprises approximately 15–20% of observations and is characterised by higher shares of residents from non-European regions, particularly Asia and Oceania, Africa, and Turkey. It exhibits the lowest average voter turnout, at approximately 51%.
2) **High-turnout, European-origin cluster.** Accounting for roughly one third of observations, this cluster is dominated by Nordic and EU-origin populations. It displays the highest average turnout, at approximately 64%, alongside lower party dominance.
3) **Mixed-origin, intermediate-turnout cluster.** The remaining cluster reflects a mixed-origin profile with intermediate turnout levels of around 62%.

Examination of cluster membership over time shows a gradual increase in the share of polling areas belonging to the European-origin, high-turnout cluster, a decline in mixed-origin areas, and a small but persistent group characterised by

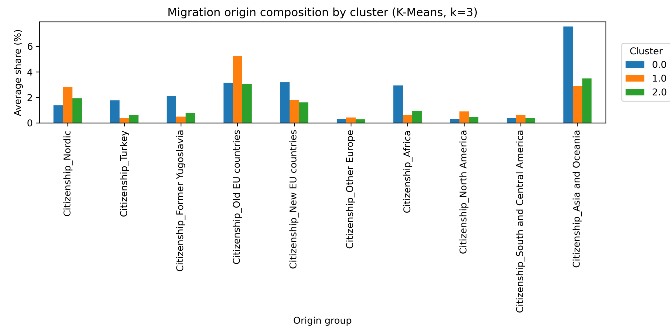

Fig. 12. Average shares of migration origin groups within each K-means cluster (k = 3). Bars represent cluster means for each origin category.

concentrated non-European origins and low turnout.

The results show a continuous association between migration composition and political behaviour: polling areas with higher non-Danish (especially non-European) population shares consistently exhibit lower turnout and distinct party support. However, changes in migration patterns are only weakly associated with changes in turnout or party votes. Thus, migration shapes long-run spatial political differences rather than short-term electoral shifts.

A. Migration and Turnout: Decision Tree Interpretation

To interpret the turnout–migration clusters, we estimated a Decision Tree classifier using citizenship composition variables.

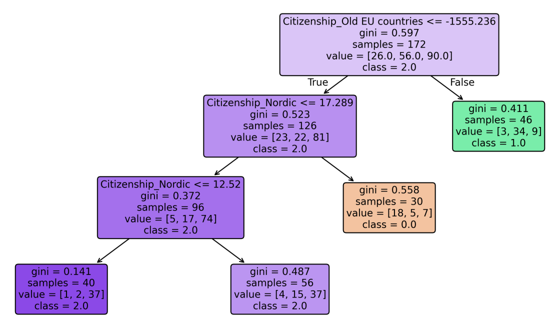

Fig. 13. Decision Tree

The Decision Tree identifies relative levels of Old EU and Nordic citizenship as the primary dimensions distinguishing clusters, indicating that European-origin composition plays a role in structuring turnout patterns. However, the model achieves only moderate predictive performance, with accuracy around 57% and balanced precision and recall across clusters. This suggests that while migration-related variables contribute to explaining turnout differences, they do not define clear-cut cluster boundaries.

Misclassifications are distributed across clusters rather than concentrated in a single group, indicating overlapping demographic profiles. In particular, one cluster is only weakly explained by citizenship composition, suggesting that its turnout behavior may be driven by factors beyond migration, such as socioeconomic characteristics or political mobilization.

Overall, the results indicate that citizenship composition is a relevant but incomplete explanation of turnout patterns, and that migration effects interact with broader contextual factors.

VIII. Shouldn't we have a final conclusion?

For swing states: Impact: Identifying swing states could inform campaign resource allocation, potentially concentrating political attention on specific areas

Appendix

A. SHAP plot

Fig. 14. SHAP beeswarm plot for the margin-based swing classifier. Points to the right (positive SHAP values) indicate features that increase the probability that an area is classified as swing, whereas points to the left indicate features that favour non-swing classifications.